



V(1st Sm.)-Computer Science-H/CC-1/CBCS

2021

COMPUTER SCIENCE — HONOURS

Paper : CC-1

(Digital Logic)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions of the following : 2×5
 - (a) State De Morgan's theorems.
 - (b) Convert $(3EA.1D)_{16} = (?)_2 = (?)_8$.
 - (c) Add $(11101.101)_2 + (1001.11)_2$ and $(6D.C)_{16} + (3B.2)_{16}$.
 - (d) Design EX-OR gate by NAND gates only.
 - (e) Subtract $(1011)_2 - (1101)_2$ using 2's complement.
 - (f) Which type of topology has been adopted to overcome race around condition? Explain.
 - (g) What are the differences between multiplexer and demultiplexer?
 - (h) How many flip-flops and other logic gates are required to design an UP decade counter?
2. (a) Simplify the logic expression $F = \sum m(0, 2, 5, 7, 8, 10, 13, 15)$ by K-map method. Design the circuit following simplified expression. Draw the truth table.
(b) Identify the maxterms from the above mentioned logic expression. Simplify it by K-map method. 7+3
3. (a) Implement $Y_{\text{difference}}$ output of a 3-bit full subtractor by the logic gates. Draw the truth table.
(b) Implement Y_{carry} output of a 3-bit full adder by NAND gates only. 6+4
4. (a) Implement $Y = \sum m(0, 4, 5, 6, 9, 10, 13, 14)$ by 8:1 multiplexer. Draw the truth table.
(b) Implement Y_{sum} output of a half adder by demultiplexer. 7+3
5. (a) Design a 2-bit multiplier by multiplying $(10)_2$ and $(11)_2$.
(b) Draw the necessary truth table. 8+2

Please Turn Over

6. (a) What is Set-Reset flip-flop? Design it by NAND gates only and explain. Draw the truth table.
(b) What is race around condition?
(c) How can the Set-Reset flip-flop be converted into D-flip-flop? Draw the truth table of D-flip-flop.
5+2+3
7. (a) Design an asynchronous UP decade counter. Explain its function.
(b) Design a MOD-5 counter showing all the count sequence.
6+4
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